



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Fall 2026)

Machine Learning

Lecture 3: Review of Calculus and Optimization

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Announcement

- Today:
 - Moved to our new classroom HU 133
 - Enroll in Gradescope ASAP if you haven't done yet
 - Participation points
 - Come to me to claim 1 point after each lecture, if
 - You asked a question, or
 - You showed/explained your solutions to in-class exercise problems
 - Maximum 3 points this semester
 - Study group registration due
 - HW 1 out
- Next Tuesday:
 - Project registration due

Recap: linear algebra review

- Vector:
 - Norm (one vector):
 - l_p norm (l_1, l_2, l_∞)
 - Distance and angle (two vectors)
 - Linear (in)dependence
 - Orthogonality: $x^T y = 0$
- Matrix:
 - Matrix-vector multiplication, matrix-matrix multiplication

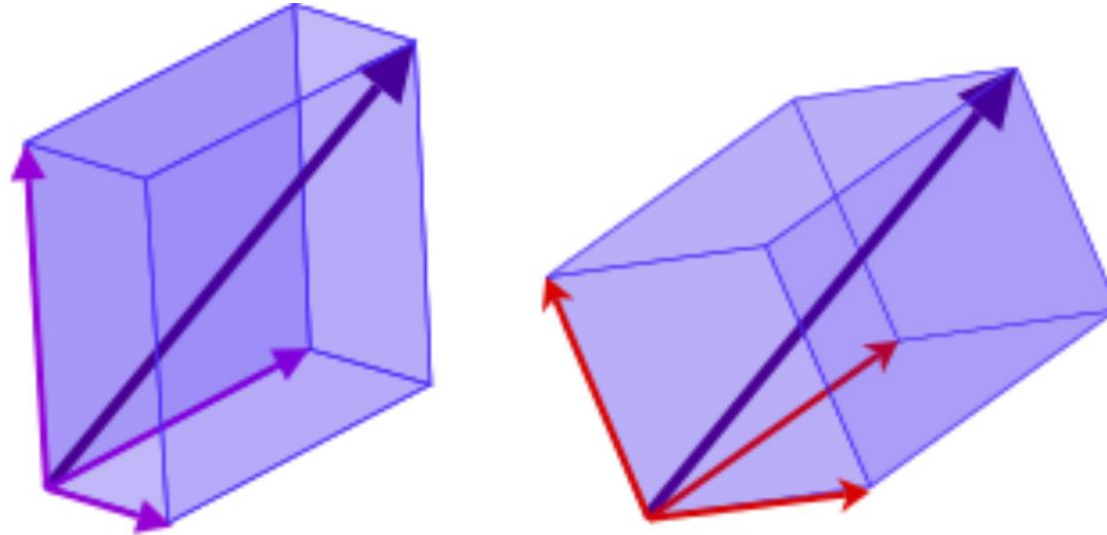
Recap: Properties of a bag of vectors: linear independence

Important to consider for machine learning algorithm design

- Given a set of vectors $\{v_1, v_2, \dots, v_n\} \in \mathbb{R}^m$, with $m \geq n$, consider the set of **linear combinations** $y = \sum_{j=1}^n \alpha_j v_j$ for arbitrary coefficients α_j 's.
- The vectors $\{v_1, v_2, \dots, v_n\}$ are **linearly independent**, if $\sum_{j=1}^n \alpha_j v_j = 0$, if and only if $\alpha_j = 0$ for all $j = 1, \dots, n$.
- Implication: if a set of vectors are linearly dependent, then one of them can be written as a linear combination of the others

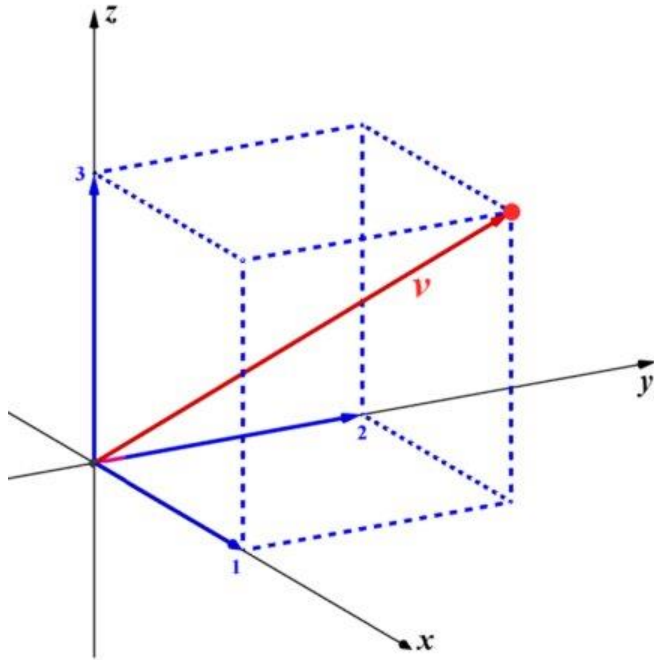
When they are linearly independent, we call this “bag” of vectors a **basis**. A basis of size m *spans* an m -dimensional vector space.

- A set of m linearly independent vectors of $\mathbb{R}^m$ is called a **basis** in $\mathbb{R}^m$: any vector in $\mathbb{R}^m$ can be expressed as a linear combination of the basis vectors.



Properties of basis

- Vectors in a basis are linearly independent.
 - Usually mutually orthogonal (Dot product of any two of them is 0).



$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Properties of a matrix

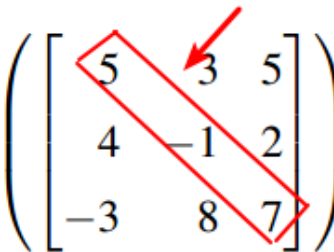
- General matrix

- Rank: max number of independent column vectors / row vectors
- Transpose: switch rows and columns

$$A \in \mathbb{R}^{m \times n} \quad A^T \in \mathbb{R}^{n \times m}$$

- Square matrix

- Trace: Sum of diagonal elements
- Determinant:


$$\text{tr} \left(\begin{bmatrix} 5 & 3 & 5 \\ 4 & -1 & 2 \\ -3 & 8 & 7 \end{bmatrix} \right) = 5 - 1 + 7 = 11.$$

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Invertible matrix

$$A^{-1} A = I$$

Orthogonal matrix

$$A^{-1} = A^T$$

Symmetric matrix

$$A^T = A$$

Eigenvalues and eigenvectors of a (square) matrix

Let A be a $n \times n$ matrix. The vector $v \neq 0$ that satisfies

$$Av = \lambda v$$

for some scalar λ is called the eigenvector of A and λ is the eigenvalue corresponding to the eigenvector v .

- ① A is symmetric, then $\lambda \in \mathbb{R}$.
- ② A is symmetric and positive semi-definite, then $\lambda \geq 0$
- ③ A is symmetric and positive definite, then $\lambda > 0$

Positive (semi)-definite matrix

Very important property for optimization, kernel methods

- A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is positive semi-definite, if and only if $x^T A x \geq 0$, for any $x \in \mathbb{R}^n$.
 - All eigenvalues of A are non-negative.
 - $X^T A X$ for any $X \in \mathbb{R}^{n \times m}$ is positive semi-definite.
- A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is positive definite, if and only if $x^T A x > 0$, for any $0 \neq x \in \mathbb{R}^n$.
 - All eigenvalues of A are positive.
 - All diagonal entries of A are positive.

In class exercise: prove $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ is a positive semi-definite matrix

- Solution 1: prove $x^T A x \geq 0$ for any vector x .
- Solution 2: prove all eigenvalues of A are all non-negative.
 - Hint: solve $\det(A - \lambda I) = 0$ to find eigenvalues.

Today's agenda

- Multi-variate calculus
 - Partial derivative and gradient
 - Chain rule
 - Multiple integrals
 - Hessian matrix
- Optimization
 - Convex set and convex function
 - Optimization problem formulation
 - Properties of convex optimization
 - Lagrange Multipliers

Multi-variate function

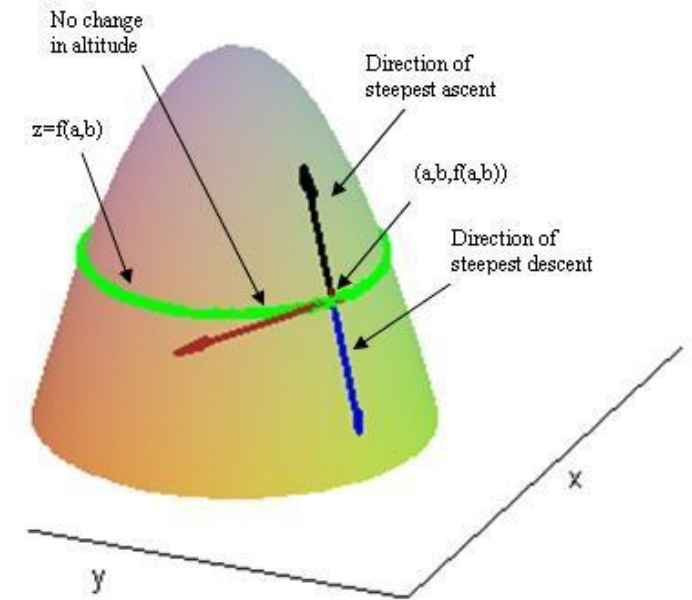
- Definition:
 - A function of two or more variables takes multiple inputs and produces a single output.
 - Examples: $f(x, y) = e^{x+y} + e^{3xy} + e^{y^4}$
- Domain:
 - Set of all possible inputs
- Range:
 - Set of possible output values.

Partial derivative

- Definition:
 - The rate of change of a function with respect to one variable, holding other variables constant.
- Notations:
 - $\frac{\partial f}{\partial x}$ or $\nabla_x f(x, y)$
- Example:
 - $f(x, y) = e^{x+y} + e^{3xy} + e^{y^4}$
 - $\frac{\partial f}{\partial x} = e^{x+y} + 3ye^{3xy}$
 - $\frac{\partial f}{\partial y} = e^{x+y} + 3xe^{3xy} + 4y^3e^{y^4}$

Gradient

- Definition:
 - A vector that points in the direction of the steepest (ascent) change.
 - Consist of multiple partial derivatives
- Example of $f(x, y)$:
 - $\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$



Chain rule

- To compute derivative of a composite function
- Example:
 - $z = f(g(t))$
 - $\frac{dz}{dt} = \frac{df}{dg} \frac{dg}{dt}$
- In-class exercise:
 - $f(x) = e^{2x}, g(x) = \sin(x)$. Find $\nabla f(g(x))$.
 - $\frac{df}{dg} = 2e^{2g(x)} = 2e^{2\sin(x)}$
 - $\frac{dz}{dt} = \frac{df}{dg} \frac{dg}{dt} = 2e^{2\sin(x)} \cos(x)$

Multiple Integrals

- Double integral: compute the volume under a surface in two dimensions.
- Example: a function $f(x, y)$ over a region R
 - $\iint_R f(x, y) dx dy$
- In-class exercise: find double integral of the function $f(x, y) = x^2 + y^2$ over $0 \leq x \leq 2$ and $1 \leq y \leq 3$.
 - $\int_0^2 x^2 dx = 8/3$
 - $\int_0^2 y^2 dx = 2y^2$
 - $\int_1^3 \left(\frac{8}{3} + 2y^2 \right) dy = 16/3 + 52/3 = 68/3$

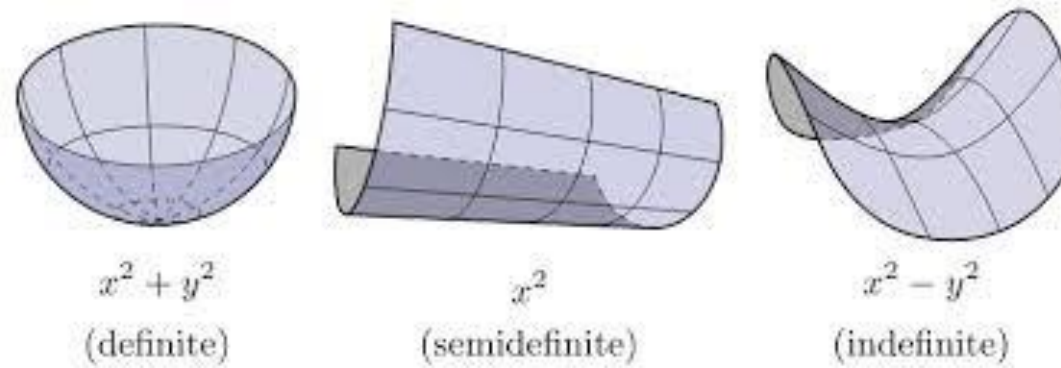
Hessian matrix – second order

$$(\mathbf{H}_f)_{i,j} = \frac{\partial^2 f}{\partial x_i \partial x_j} \quad \mathbf{H}_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

- Quadratic approximation of a function
 - $f(x + y) \approx f(x) + y^T \nabla f(x) + \frac{1}{2} y^T \nabla^2 f(x) y$

Hessian matrix – second order

- Hessian matrix is symmetric
- Hessian matrix and local curvature of the function
 - Minimum: Hessian is positive definite
 - Maximum: Hessian is negative definite
 - Saddle point: Hessian is indefinite (not positive/negative definite)



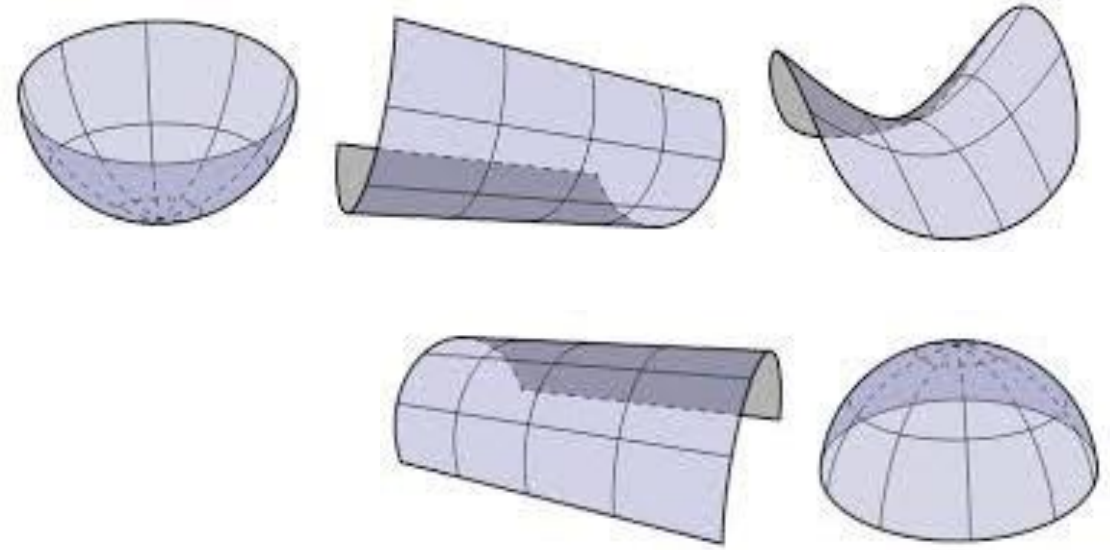
Quadratic Function

- $f(x) = \frac{1}{2}x^T Ax + b^T x + c$

- Gradient: $\nabla f(x) = Ax + b$
- Hessian: $\nabla^2 f(x) = A$

- Quadratic programming:

- $\min f(x) = \frac{1}{2}x^T Ax + b^T x + c$
- Key: check Hessian matrix!
 - Hessian is positive (semi)definite: minimum (local or global)
 - Hessian is negative (semi)definite: maximum (local or global)
 - Hessian is indefinite: undetermined, changing curvature

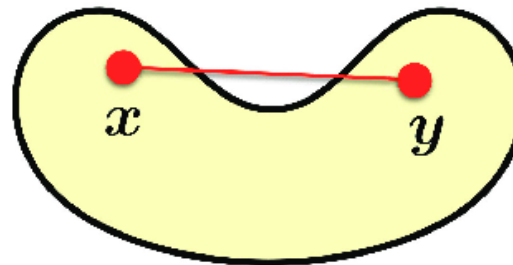
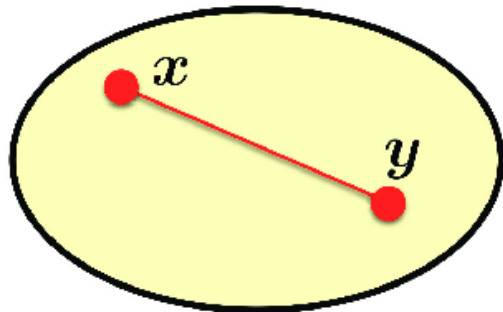


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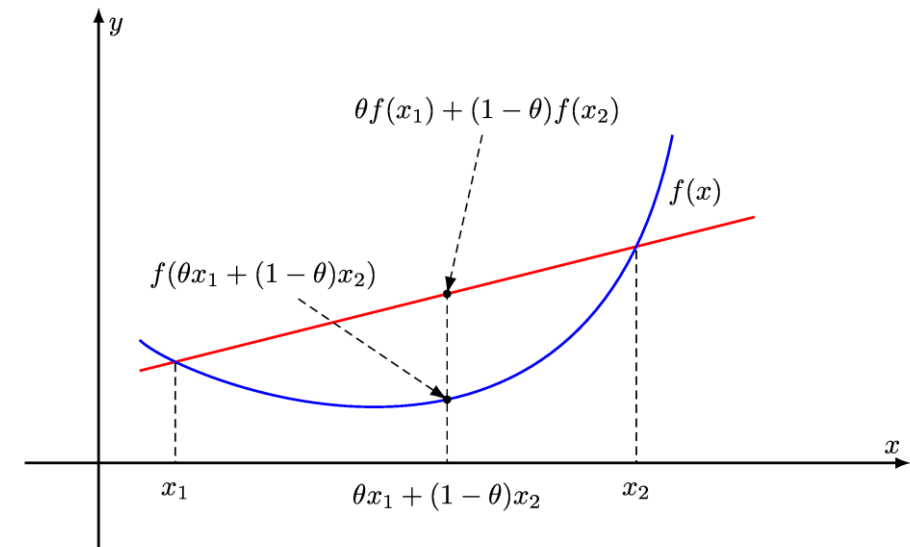
Convex Sets

- Definition:
 - A set $C \subseteq R^n$ is convex if for any two points $x_1, x_2 \in C$, $\theta x_1 + (1 - \theta)x_2 \in C$ for all $\theta \in [0,1]$.
- Interpretation:
 - A set $C \subseteq R^n$ is convex if, for any two points $x_1, x_2 \in C$, the line segment connecting them is also entirely within C .
- Discussion: are they convex sets?
 - (1) $[0,1]$
 - (2-3)



Convex functions

- Definition:
 - A function $f: C \rightarrow R$ is convex if C is a convex set and for all $x_1, x_2 \in C$ and $\theta \in [0,1]$:
 - $f(\theta x_1 + (1 - \theta)x_2) \leq \theta f(x_1) + (1 - \theta)f(x_2)$
- Interpretation:
 - A convex function lies below the line segment connecting any two points on its graph.
- Discussion: propose some convex functions
 - Example: linear functions, quadratic functions, exponential functions.



Optimization problem formulation

- $\min_{x \in X} f(x)$
 - Given $x \in X$, find the minimum value of $f(x)$
- $\operatorname{argmin}_{x \in X} f(x)$
 - Given $x \in X$, find the x such that $f(x)$ is minimized
- In-class exercise:
 - What's $\min_{x \in [2,3]} x^2$?
 - What's $\operatorname{argmin}_{x \in [2,3]} x^2$?

Convex optimization problem formulation

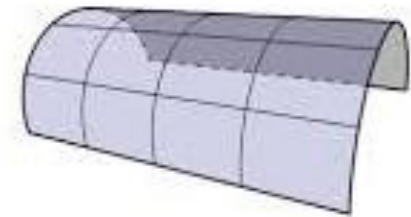
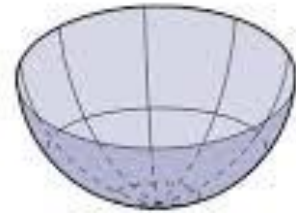
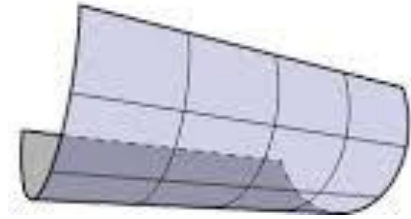
- $\min f(x),$
 - s. t. $g(x) \leq 0, h(x) = 0.$
-
- $f(x)$ is the convex objective function
 - $g(x)$ is convex inequality constraint
 - $h(x)$ is equality constraint

1-dimensional optimization

- $f(x) = x^3 + 3x^2 - 24x + 2$
 - First, solve $f'(x) = 0$ to get all solutions $f'(x) = 3x^2 + 6x - 24 = 0, x_1 = -4, x_2 = 2$.
 - Second, for each solution, check $f''(x)$: $f''(x) = 6x + 6$
 - $f''(x) > 0$: minimum (local or global) $x = 2$
 - $f''(x) < 0$: maximum (local or global) $x = -4$
 - $f''(x) = 0$: undetermined, changing curvature

Hessian matrix and convex function

- $\nabla^2 f(x) \succcurlyeq 0$, then $f(x)$ is convex
 - No local minimum
- $\nabla^2 f(x) \succ 0$, then $f(x)$ is strongly convex
 - Unique global minimum
- $-\nabla^2 f(x) \succcurlyeq 0$, then $f(x)$ is concave
 - No local maximum
- $-\nabla^2 f(x) \succ 0$, then $f(x)$ is strongly concave
 - Unique global maximum

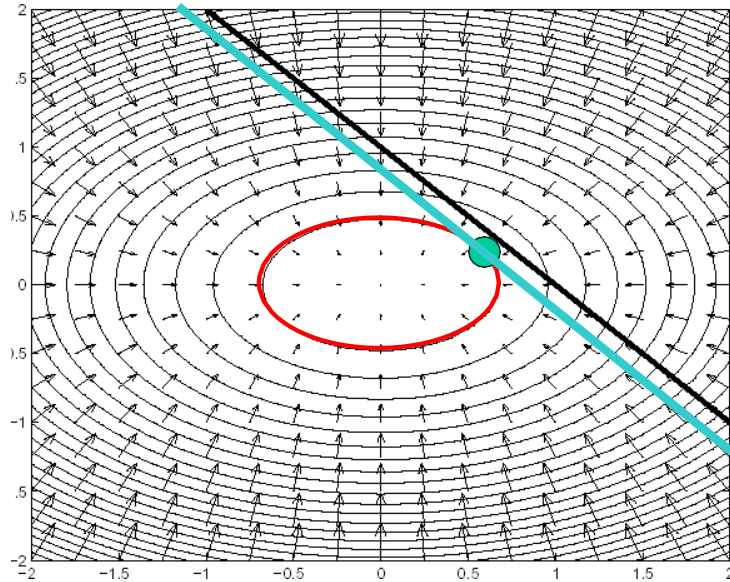


Lagrange multipliers to handle constraints

- The Lagrangian function combines the objective function with the constraints using multipliers.
- Example: $\max xy, \text{ s. t. } x + y = c$
 - Solution 1: use $y = c - x$, then objective problem is $\max x(c - x)$, so $x = y = c/2$ is the optimal solution.
 - Solution 2 (Lagrange multiplier):
 - $L(x, y, \lambda) = xy - \lambda(x + y - c)$
 - Differentiate with regards to x and y , we have $x = y = \lambda$

Equality constrained problem

- $\min f(x, y) = x^2 + 2y^2 - 2$
- s.t. $x + y = 1$



Equality constrained problem

- $\min f(x, y) = x^2 + 2y^2 - 2$
- s.t. $x + y = 1$

Introduce Lagrangian multiplier λ and form

- Solution: $L(x, y, \lambda) = x^2 + 2y^2 - 2 - \lambda(x + y - 1)$

Then, differentiate with respect to x, y, λ : and set derivative to 0.

$$\left. \begin{aligned} \frac{\partial L}{\partial x} &= 2x - \lambda = 0 \quad \Rightarrow \quad \lambda = 2x \\ \frac{\partial L}{\partial y} &= 2y - \lambda = 0 \quad \Rightarrow \quad \lambda = 4y \\ \frac{\partial L}{\partial \lambda} &= -x - y + 1 = 0 \quad \Rightarrow \quad -x - y + 1 = 0 \end{aligned} \right\} \begin{aligned} \lambda &= \frac{4}{3} \\ x &= \frac{2}{3} \\ y &= \frac{1}{3} \end{aligned}$$